

Project ATHENA — Day 6

Theme: Moving Averages — Your First Quantitative Signal

"A moving average doesn't predict the future. It helps you see the present more clearly."

Objective: Understand moving averages from first principles, implement them in Python, and build the first reusable indicator in ATHENA.

Time Required: ~40 minutes

Why Today's Lesson Matters

So far you've learned:

- How markets work
- Market participants
- Risk management
- Data architecture
- Returns and volatility

Today, we're building the **first brick of the strategy engine**.

Almost every quantitative system—from simple trend-following models to sophisticated machine learning pipelines—uses some form of smoothing. Moving averages are one of the simplest and most widely used ways to reduce market noise.

The Problem: Markets Are Noisy

Imagine tracking Bengaluru's daily temperature.

30°

31°

29°

32°

30°

31°

29°

Is the weather warming up?

It's hard to tell because of day-to-day fluctuations.

Stock prices behave the same way.

100

101

99

102

101

103

100

If you react to every price change, you'll constantly change your opinion.

A moving average smooths out those fluctuations so you can focus on the broader direction.

What Is a Moving Average?

A moving average is simply the average of the most recent N prices.

Example:

Last 5 closing prices:

100

102

101

103

104

5-day SMA:

$$(100 + 102 + 101 + 103 + 104) / 5$$

$$= 102$$

Tomorrow:

102

101

103

104

105

The oldest value drops out.

The newest value enters.

The average "moves."

Simple Moving Average (SMA)

Every observation gets equal weight.

Price:

100

101

102

103

104

↓

Average = 102

Advantages:

- Easy to understand.
- Stable.
- Good for identifying medium- and long-term trends.

Disadvantages:

- Slow to react when prices change rapidly.
-

Exponential Moving Average (EMA)

EMA gives **more importance to recent prices**.

Recent Prices

↓

Higher Weight

↓

EMA

Advantages:

- Responds more quickly to trend changes.
- Popular for shorter-term systems.

Disadvantages:

- Can react to temporary noise more than SMA.

When Should You Use Each?

As a rule of thumb:

- **SMA**: better when you care about stable long-term trends.
- **EMA**: better when you want faster reactions.

Neither is "better." They answer different questions.

A useful mindset is:

Indicators are tools, not truths.

The Biggest Beginner Mistake

Many traders ask:

"Which moving average is best?"

The better question is:

"For this strategy, does any moving average improve performance compared to doing nothing?"

That's a research question.

And research is what ATHENA is built to support.

Engineering Perspective

A moving average is not a strategy.

It is a **feature**.

Think back to your ML projects.

You never trained a model directly on raw text—you engineered features.

Similarly:

Raw Prices



Moving Average



Momentum



Volatility



Strategy

Moving averages are just one of many features.

Practical Exercise (~40 Minutes)

Part 1 — Implement ATHENA's First Indicator (25 min)

Create:

src/indicators/moving_average.py

Implement:

```
def sma(prices, window):
```

```
    """
```

```
    Return the Simple Moving Average.
```

```
    """
```

```
    pass
```

```
def ema(prices, window):
```

```
    """
```

```
    Return the Exponential Moving Average.
```

```
    """
```

```
    pass
```

Use **pandas** rather than writing loops.

One of your design goals should be readability and reliability.

Part 2 — Verify with a Small Dataset (10 min)

Create a notebook:

notebooks/

moving_average_demo.ipynb

Generate a simple synthetic price series:

100

101

102

101

103

104

105

106

105

107

Compute:

- SMA(5)
- EMA(5)

Observe:

- Which reacts faster?
- How much smoother is the SMA?

You don't need charts yet—printing the results is enough.

Part 3 — Update the Architecture (5 min)

ATHENA now contains:

src/

```
|— ingestion/
|— cleaning/
|— indicators/
|   └─ moving_average.py
|— features/
|— performance/
|— portfolio/
|— risk/
|— scanner/
|— strategies/
└─ backtesting/
```

The important point isn't the file itself—it's the separation of responsibilities.

Engineering Principle of the Day

Never let strategy code calculate indicators directly.

Instead:

Indicators



Return Values



Strategy Consumes Values

This keeps your code modular.

Later, one strategy might use SMA.

Another might use EMA.

A third might use neither.

No duplication.

Review Question

Suppose two traders use the same strategy.

Trader A uses a 10-day SMA.

Trader B uses a 10-day EMA.

A stock suddenly rises sharply over two days.

Which indicator is more likely to react first?

Answer: The **EMA**, because it assigns greater weight to recent prices.